1. Primary Alignment: Items 5 and 7

- **Item 5: Understand and utilise constructs based on definitions, rules and formal systems...**
  - **Analysis:** This is the foundation of the solution. Students must apply a fundamental geometric rule: a full circle measures 360°. The problem explicitly states the space is divided into "three equal sectors," requiring the student to use this formal system (angular measurement) within the specific context of the door's rotation.
- **Item 7: Manipulate numbers... and geometric representations**
  - **Analysis:** Students interact with a geometric representation (the plan view of the revolving door). The solving process involves manipulating this representation—breaking a total rotation of 360° into three equal parts (division by 3)—demonstrating mathematical operation on a geometric form.

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2. Significant Alignment: Items 10 and 1

- **Item 10: Make sense of, relate and use a variety of representations when interacting with a problem**
  - **Analysis:** The problem combines a textual description ("three wings," "three equal sectors") with a visual diagram showing the door's structure (p. 1). Students must relate these two representations to extract the key mathematical information: that the wings form central angles of a circle.
- **Item 1: Perform a simple calculation**
  - **Analysis:** The final step is a direct and simple arithmetic operation: $360^\circ \div 3 = 120^\circ$. This is a classic example of performing a single-step calculation to reach a mathematical solution.

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3. Secondary Alignment: Item 13

- **Item 13: Use an understanding of the context to guide or expedite the mathematical solving process**
  - **Analysis:** An understanding of the real-world context (a revolving door) helps the student quickly realize this is a problem about dividing a circular rotation. While some contextual data (like the 2-meter diameter) is irrelevant for calculating the angle, understanding the context allows the student to filter out unnecessary information and focus on the relevant geometric properties.

1. 核心关联：第 5 项与 第 7 项

- **第 5 项：Understand and utilise constructs based on definitions, rules and formal systems...**
  - **分析：** 这是解题的基础。学生必须运用几何学的基本定义和规则：即一个圆周的总度数为 360°。题目明确指出空间被分为“三个相等扇区”，学生需要利用这一规则将抽象的几何定义（周角）应用于特定情境 (p. 1)。
- **第 7 项：Manipulate numbers... and geometric representations**
  - **分析：** 学生需要处理几何表征（旋转门的俯视图）。解题过程涉及对几何图形的拆解——将一个完整的圆（360°）处理为三个相等的部分（除以 3），这体现了对几何表现形式的数学操作能力 (p. 1)。

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2. 重要关联：第 10 项与 第 1 项

- **第 10 项：Make sense of, relate and use a variety of representations when interacting with a problem**
  - **分析：** 题目结合了文字描述（“three wings”、“three equal sectors”）和视觉图示（显示门翼结构的平面图）。学生必须理解这两种表征之间的关系，并从图示中提取出“圆心角”这一关键数学信息 (p. 1)。
- **第 1 项：Perform a simple calculation**
  - **分析：** 最终的计算步骤非常直接：$360^\circ \div 3 = 120^\circ$。这是一个典型的简单计算过程，旨在得出最终的数学解 (p. 1)。

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3. 次要关联：第 13 项

- **第 13 项：Use an understanding of the context to guide or expedite the mathematical solving process**
  - **分析：** 对现实背景（旋转门）的理解有助于学生快速意识到这是一个关于圆周平分的问题。背景信息（如直径 2 米）虽然在计算角度时是干扰项，但理解情境能帮助学生筛选出对解决特定问题（角度计算）有用的信息。